

Syllabus

Spring semester 2026

Course Title	Special Topics in Chemistry (Sensor Chemistry)		Course Code		CB5103		Credit	3	Classroom	3305
			Type		Elective		Hours	3		
Teacher	Name	Class no.	Office	Office Hour*		Contact		E-mail		
	Park, Seonhwa	1	5604	M3, Tu3, F3		378		spark1122@ksa.kaist.ac.kr		
	Park, Seonhwa	2	5604	M3, Tu3, F3		378		spark1122@ksa.kaist.ac.kr		

*Office Hour: Time for students to ask questions and discuss lecture or lab topics with teachers for deeper understanding.

1. Course Description

This course explores the chemical and biological principles of sensor technologies, emphasizing electrochemical, optical, and fluorescent biosensing mechanisms. Students will study molecular recognition, signal transduction, and analytical performance in enzyme-, antibody-, and nucleic-acid-based biosensors.

- Understand the fundamentals of chemical and biological sensing principles.
- explain the signal conversion mechanisms of electrochemical, optical and fluorescence-based sensors.
- compare the structure and sensing processes of enzyme-, antibody-, and DNA-based biosensors.
- Analyze performance metrics such as sensitivity, selectivity and limit of detection of sensors.
- evaluate the applications of the latest sensor technologies (wearable, paper-based, smart sensors).

2. Core Target Competencies

Subject Core Competencies				
Knowledge Comprehension	Analytical and Creative Thinking	Complex Problem-Solving Skills	Complex Communication	Leadership and Teamwork
Students will learn the principles and applications of chemical sensors and biosensors through lectures, self-directed learning, and in-class group discussions. The content covered in this course is broadly divided into three areas: (1) chemical sensors, (2) biosensors, and (3) chemical signal generation and detection methods. (1) Chemical Sensors - Gas sensors, environmental sensors, pH sensors (2) Biosensors - Enzyme-based sensors, immunosensors, DNA sensors (3) Chemical Signal Generation and Detection Methods - Electrochemical, optical, and fluorescence-based signals				

3. Textbooks and References

A. Textbook: Brian R. Eggins, *Chemical Sensors and Biosensors*, 2nd ed., Wiley, 2002.

B. References

- Dermot Diamond, *Principles of Chemical and Biological Sensors*, Wiley, 1998.
- Latest review articles (*Analytical Chemistry, Biosensors & Bioelectronics*, and etc.)

4. Class Method

- A. Lecture
- B. Presentation and discussion

5. Evaluation

A. Evaluation Criteria

Workbook	Presentation	Midterm exam	Attendance and Attitude	Sum
20	30	40	10	100

B. Evaluation Method

(1) Midterm Exams Criteria

- Exams will include multiple-choice, short-answer, and essay questions. The exam will be conducted in written format.

(2) Presentation

- Minimize the use of textbook content and focus primarily on supplementary materials. Select a topic of interest from recent research on chemical sensors or biosensors, explore it in depth, and explain your understanding in a manner that persuades and engages your peers. Presentations will be assessed not only for accuracy and completeness, but also for relevance to the chapter, their effectiveness in motivating peers to learn, and the level of active participation in other students' presentations (attitude of attentive listening and active questioning).

(3) Workbook Criteria

- A workbook is provided for use in self-study and class activities. You must complete the workbook assignments for the class before the next class. You can also use the workbook to develop problem-solving skills and expand your understanding of concepts.

(4) Learning Attitude and Attendance Evaluation Criteria

- Engaging in unrelated activities such as gaming or non-class-related use of computers or smartphones four or more times during the semester will result in an F grade.
- A deduction of 3 points will be applied for absences without a valid reason.

6. Criteria for Awarding Credits

- ☞ Within the same course, up to 70% of students will receive A or B grades regardless of their scores or class placement. However, only when the score differences among students are not significant may grades of A or B be awarded to 70% or more of the students.
- ☞ Students scoring 50 points or below will receive an F grade.
- ☞ Students who are absent three or more times without a valid reason will receive an F grade.
- ☞ If you are uncooperative in presentations and activities, you may receive an F grade depending on

the circumstances.

☞ **Students with excellent academic achievement are recommended for the Academic Excellence Award and the Principal's Award, and are recorded according to core competencies in the Subject Details. However, students with poor academic attitude or poor attendance are excluded.**

- Attendance for at least three-quarters of the total class hours is required to earn credit.
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- Calculation of the basis score for awarding credit to absentees : For officially excused absences or medically excused absences, the recognized score for a missed exam will be calculated as follows: (Percentile rank equivalent to the total score obtained in presentation and workbook of the same course taken by the student) × (100% for officially excused absences; 80% for medically excused absences).
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- If a basis score cannot be calculated, the grading method : If a basis score cannot be calculated, grading will follow the Academic Performance Management Regulations.
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- ※ Refer to the "Academic Performance Management Regulations" for the percentage of credit points applied.

7. Weekly Plans

Week (period)	Topic and Contents	(Notes)
Week 1 2.23.~3.1.	Introduction to Sensors : Definition, classification, basic components, overview of chemical vs. biosensors.	
Core Competencies	◆ Knowledge Comprehension, Analytical and Creative Thinking, Complex Problem-Solving Skills	
Week 2 3.2.~3.8.	Principles of Recognition : Chemical and biological recognition, binding affinity, enzyme/antibody/nucleic-acid interactions.	3.2. Substitute Holiday
Core Competencies	◆ Knowledge Comprehension, Analytical and Creative Thinking, Complex Problem-Solving Skills	
Week 3 3.9.~3.15.	Optical and Fluorescent Fundamentals I : Optical Interactions and Absorbance-Based Phenomena	
Core Competencies	◆ Knowledge Comprehension, Analytical and Creative Thinking, Complex Problem-Solving Skills	
Week 4 3.16.~3.22.	Optical and Fluorescent Fundamentals II : Fluorescence, Quenching, Enhancement, and FRET	
Core Competencies	◆ Knowledge Comprehension, Analytical and Creative Thinking, Complex Problem-Solving Skills	
Week 5 3.23.~3.29.	Electrochemical Fundamentals I : Redox reactions, electrode potentials, Nernst equation.	
Core Competencies	◆ Knowledge Comprehension, Analytical and Creative Thinking, Complex Problem-Solving Skills	
Week 6 3.30.~4.5.	Electrochemical Fundamentals II : Cyclic voltammetry, differential pulse voltammetry, impedance spectroscopy.	
Core Competencies	◆ Knowledge Comprehension, Analytical and Creative Thinking, Complex Problem-Solving Skills	
Week 7 4.6.~4.12.	Surface Modification and Bioconjugation : Self-assembled monolayers, nanomaterials, biomolecule immobilization.	
Core Competencies	◆ Knowledge Comprehension, Analytical and Creative Thinking, Complex Problem-Solving Skills	
Week 8 4.13.~4.19.	Midterm Exams (4.14.~4.17.)	4.13. Regular Classes

Week 9 4.20.~4.26.	Practical sensors using optical and fluorescence signal : hCG and COVID-19 detection using later flow strip	
Core Competencies	◆ Knowledge Comprehension, Complex Communication, Leadership and Teamwork	
Week 10 4.27.~5.3.	Practical sensors using electrochemical signal : Cellulose strip sensors, sweat and saliva analysis, smartphone-integrated detection platforms	
Core Competencies	◆ Knowledge Comprehension, Complex Communication, Leadership and Teamwork	
Week 11 5.4.~5.10.	Chemical and bio- sensors using optical signal : Environmental and diagnostic applications	5.5. Children's Day
Core Competencies	◆ Knowledge Comprehension, Complex Communication, Leadership and Teamwork	
Week 12 5.11.~5.17.	Chemical and bio- sensors using fluorescence signal : Environmental and diagnostic applications	
Core Competencies	◆ Knowledge Comprehension, Complex Communication, Leadership and Teamwork	
Week 13 5.18.~5.24.	Chemical sensors using electrochemical signal : Environmental and diagnostic applications	
Core Competencies	◆ Knowledge Comprehension, Complex Communication, Leadership and Teamwork	
Week 14 5.25.~5.31.	Biosensors using electrochemical signal 1 : Environmental and diagnostic applications	5.25. Substitute Holiday
Core Competencies	◆ Knowledge Comprehension, Complex Communication, Leadership and Teamwork	
Week 15 6.1.~6.7.	Biosensors using electrochemical signal 2 : Environmental and diagnostic applications	6.3. Local Election Day
Core Competencies	◆ Knowledge Comprehension, Complex Communication, Leadership and Teamwork	
Week 16 6.8.~6.14.	Make-up class week	
Week 17 6.15.~6.19.	Final Exams (6.18.~6.19.)	

※ The above plans may change according to the academic schedule.